2 Polynomials

- A polynomial is an algebraic expression composed of variables, coefficients and non-negative integer exponents, combined using addition, subtraction and multiplication.
- **Examples**

Polynomial	Not a Polynomial
$3x^3 - 2x^2 + 7$	$\frac{1}{x} + 5$
$2x$	$\sqrt{x} + 2$
$3a + 6 + 4$	$4 + \frac{1}{x}$

Algebraic Expression

- An algebraic expression is a combination of → Constants + Variables + Operations [+, −, ×, ÷]
- Example : $2x + 3y$, $x^2 - 5x$

2.1 Polynomial in One Variable

- If a polynomial contains only one variable then it is called polynomial in one variable.
- **Examples** $2x + 3$, $6x^2 - 5x + 2$, $8y + 6$
- A polynomial in one variable x is an expression of the form :

$$p(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_{n-1} x^{n-1} + a_n x^n$$

Where, $a_0, a_1, a_2, \ldots, a_{n-1}, a_n$ are constants (real numbers).
$a_n \neq 0$ and n is a non-negative integer.

- The constants $a_0, a_1, a_2, \ldots, a_n$ are also known as coefficients of polynomial.

Degree of a Polynomial

- The highest power of the variable in a polynomial is known as the degree of the polynomial.
- **Examples** (a) $3x + 5 \rightarrow$ degree = 1 (b) $4x^2 - 7x + 2 \rightarrow$ degree = 2
(c) $x^3 + 2x - 9 \rightarrow$ degree = 3 (d) $x^4 + 3x^3 - 4 \rightarrow$ degree = 4

Types of Polynomial

Polynomials are divided into two types → **on the basis of degree** and **on the basis of terms**

Types of Polynomial on the Basis of Degree

Linear Polynomial
- Polynomial of degree 1
- **Example** $2x + 3$
- General form :
  $p(x) = ax + b, a \neq 0$

Quadratic Polynomial
- Polynomial of degree 2
- **Example** $x^2 + 5x + 6$
- General form :
  $p(x) = ax^2 + bx + c, a \neq 0$

Cubic Polynomial
- Polynomial of degree 3
- **Example** $x^3 - x^2 + 4$
- General form :
  $p(x) = ax^3 + bx^2 + cx + d, a \neq 0$

Biquadratic Polynomial
- Polynomial of degree 4
- **Example**
  $x^4 + 2x^2 - 1$

Constant Polynomial
- Only constant term
- **Example**
  $4, 6, -4, -8, 100...$
  ↓
  $4x^0, 3x^0, 5x^0....$

Zero Polynomial
- Degree not defined
- **Examples**
  $0x^0, 0x^2, 0x^{100}....$
- **General form**
  $p(x) = 0$

Types of Polynomial on the Basis of Terms

Monomial
- Containing 1 term only (Mono means one)
- **Example** or
  $5x, -7y^2, 9, 8x^2$

Binomials
- Containing two non-zero terms (Bi means two)
- **Example** or
  $8 + 3x, x + 3x^2, 4 + x^3$

Trinomials
- Containing three non-zero terms (Tri means three)
- **Example** or
  $x^2 + x + 1, 2x^2 - 5x + 3$

Value of a Polynomial

- The number you get after replacing the variable (say x) with a specific real number and performing the calculation.
- The value of a polynomial $p(x)$ at $x = a$ is written as $p(a)$.

- **Example** If $p(x) = 4x^2 - 2x + 3$

  $p(-1) = 4(-1)^2 - 2(-1) + 3 = 4 + 2 + 3 = 9$

$\therefore$ The value of $p(x)$ at $x = -1$ is 9.

Zeroes of a Polynomial

- If $p(a) = 0$, then a is a zero of polynomial $p(x)$.
- **Examples** $p(x) = x + 3$

  For zero of $p(x)$, put $p(x) = 0$

  $\Rightarrow x + 3 = 0 \Rightarrow x = -3$

  so -3 is a zero of $p(x)$. and $p(x) = x^2 - 9$

  For zero of $p(x)$, put $p(x) = 0$

  $\Rightarrow x^2 - 9 = 0 \Rightarrow x^2 = 9$

  $\boxed{x = 3, -3}$

  So, 3 and -3 are the zeros of polynomial.
- Geometrical meaning of zeroes of a polynomial $\rightarrow$

  Zeroes of polynomial = points where graph of $y = p(x)$ cuts the X-axis.

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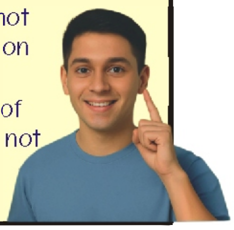
- A constant polynomial (like 5) has no zero.
- Every linear polynomial (degree 1) has exactly one zero.
- Zero may or may not be a root depends on the polynomial.
- Number of zeroes of the polynomial can not exceed its degree.



2.2 Graph of Linear Polynomial (Degree 1)

- **Example** $4x + 2$

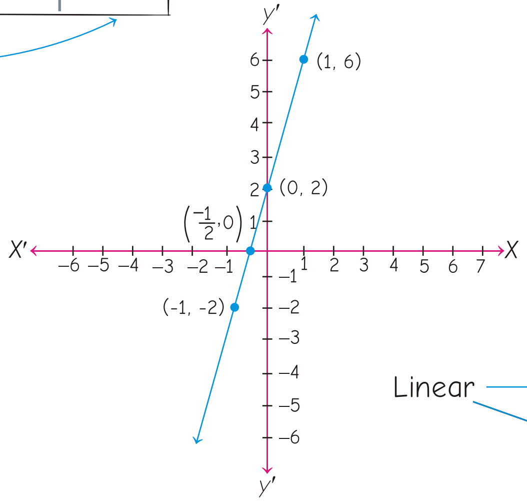
  Let $y = 4x + 2$

To draw its graph, we need two or more than two values of y corresponding to the different values of x. For this consider the following table.

x	0	1	-1	$-\frac{1}{2}$
y	2	6	-2	0

p (x)

$-\frac{1}{2}$ is a zero



Linear $\rightarrow$ Graph is a straight line

$\rightarrow$ cuts X-axis at exactly 1 point (so 1 zero)

2.3 Graph of Quadratic Polynomial (Degree 2)

Quadratic → Graph is parabola

Quadratic → It cuts X-axis at Two distinct points → 2 zeroes

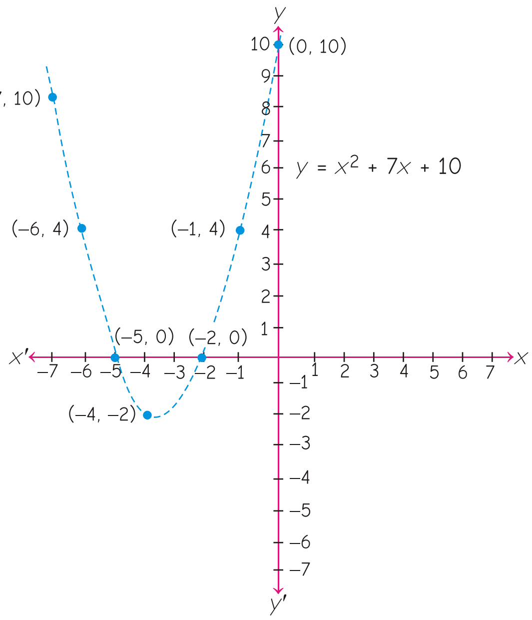
- Exactly one point → 1 zero
- No point → 0 zero

- **Example** $x^2 + 7x + 10$

Let $y = x^2 + 7x + 10$

For this, consider the following table.

x	–2	–1	0	–4	–5	–6	–7
y	0	4	10	–2	0	4	10

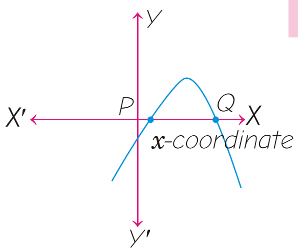


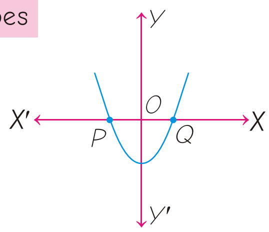
A linear polynomial can have atmost 1 zero and a quadratic polynomial can have atmost 2 zeroes.

Hence, –2 and –5 are zeroes of the given quadratic polynomial.

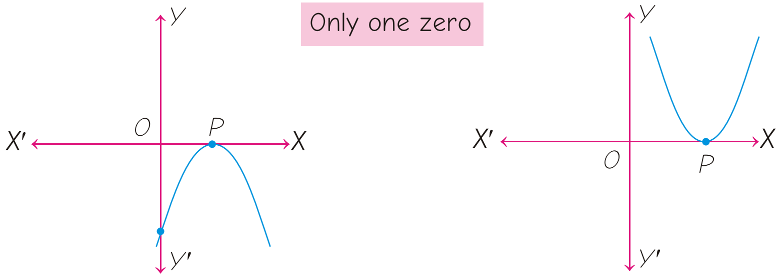
Case (i) **The graph cuts X-axis at two distinct points P and Q.**

2 Zeroes

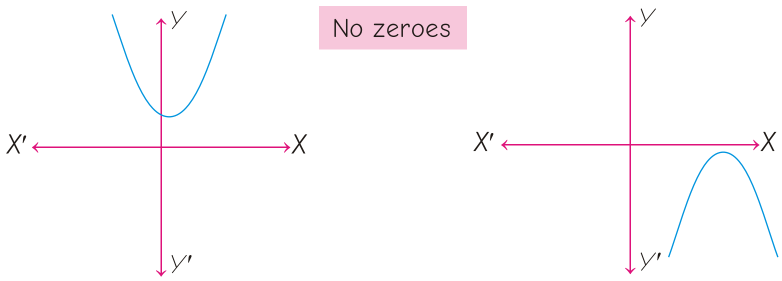




Case (ii) **The graph cuts X-axis at exactly one point**



Case (iii) **The graph is either completely above X-axis or completely below the X-axis. So, it does not cut the X-axis at any point.**



- A polynomial $p(x)$ of degree n the graph of $y = p(n)$ intersects the X-axis atmost n points.
- Hence, a polynomial $p(x)$ of degree n can have atmost n zeroes.

2.4 Middle Term Splitting Method (Factorisation)

- Used to find zeroes of a quadratic polynomial.
- **Steps**

Write quadratic polynomial : $ax^2 + bx + c$

↓

Multiply $a \times c$

↓

Find two numbers whose : Product $= a \times c$ and sum $= b$

↓

Split the middle term using these numbers.

↓

Take common terms and factorise.

$ax^2 + bx + c$ → Product $= ac$

→ Sum $= b$

$+bx$ → Middle term

Example 1

$x^2 - 8x + 12$

Product $= 12$, sum $= -8$

The terms $6x$ and $2x$ make $12x^2$ and $6x + 2x = 8x$

$\boxed{6,\ 2}$

$$x^2 - 8x + 12 = 0$$

$$x^2 - (6+2)x + 12 = 0$$

$$\boxed{x^2 - 6x} - \boxed{2x + 12} = 0$$

$$x(x-6) - 2(x-6) = 0$$

$$(x-6)(x-2) = 0$$

$$x - 6 = 0 \qquad x - 2 = 0$$

$$x = 6 \qquad x = 2$$

Example 2

$x^2 - 2x - 8$

Product $= -8$ and sum $= -2$

The terms $2x$ and $4x$ makes $8x^2$ and $2x - 4x = -2x$

$\boxed{2,\ 4}$

$$x^2 - 2x - 8 = 0$$

$$x^2 - (4-2)x - 8 = 0$$

$$\boxed{x^2 - 4x} + \boxed{2x - 8} = 0$$

$$x(x-4) + 2(x-4) = 0$$

$$(x-4)(x+2) = 0$$

$$\boxed{x = 4, -2}$$

2.5 Relationship between the Zeroes and Coefficient of Quadratic Polynomial

For a Linear Polynomial ($ax + b$, $a \neq 0$)

$$\text{Zero} = -\frac{b}{a} = -\frac{\text{Constant term}}{\text{Coefficient of } x}$$

For a Quadratic Polynomial ($ax^2 + bx + c, a \neq 0$)

Sum of zeroes, $\alpha + \beta = -\frac{b}{a} = -\frac{\text{Coefficient of } x}{\text{Coefficient of } x^2}$

Product of zeroes, $\alpha\beta = \frac{c}{a} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$

Formation of Quadratic Polynomials

If α and β are zeroes of a quadratic polynomials, then quadratic polynomial will be x^2 – (Sum of zeroes)x + Product of zeroes

i.e. $x^2 - (\alpha + \beta)x + \alpha\beta$.

Example Find a quadratic polynomial whose zeroes are 6 and –3.

Sol. Let $\alpha = 6$ and $\beta = -3$

$\therefore\ \alpha + \beta = 6 + (-3) = 3$ and $\alpha\beta = 6 \times (-3) = -18$

So, quadratic polynomial is given by

$x^2 - (\alpha + \beta)x + \alpha\beta = x^2 - 3x - 18$.

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Ques. 1 If $p(x) = cx + d$, then find the zero of $p(x)$

Sol. Given, $p(x) = cx + d$

For zero of $p(x)$, put $p(x) = 0$

$\Rightarrow\quad cx + d = 0$

$\Rightarrow\quad cx = -d$

$\Rightarrow\quad x = \frac{-d}{c}$

Check $p\left(-\frac{d}{c}\right) = c\left(-\frac{d}{c}\right) + d = -d + d = 0$

So, the zero of $p(x)$ is $-\frac{d}{c}$.

Ques. 2 If 2 is a zero of polynomial $f(x) = ax^2 - 3(a-1)x - 1$, then find the value of a.

Sol. Given polynomial is $f(x) = ax^2 - 3(a-1)x - 1$

Since, 2 is a zero of polynomial.

$\therefore\ f(2) = 0 \Rightarrow a(2)^2 - 3(a-1)2 - 1 = 0$

$\Rightarrow\quad 4a - 6a + 6 - 1 = 0$

$\Rightarrow\quad -2a = -5$

$\therefore\quad a = \frac{5}{2}$

Ques. 3 Find the zeroes of the quadratic polynomial $4s^2 - 4s + 1$.

Sol. Let $p(s) = 4s^2 - 4s + 1$

For zeroes of $p(s)$, put $p(s) = 0$

$\Rightarrow \quad 4s^2 - 4s + 1 = 0$

$\Rightarrow \quad 4s^2 - (2+2)s + 1 = 0$

$\Rightarrow \quad 4s^2 - 2s - 2s + 1 = 0$

$\Rightarrow \quad 2s(2s-1) - 1(2s-1) = 0$

$\Rightarrow \quad (2s-1)(2s-1) = 0$

$\Rightarrow \quad 2s - 1 = 0, \qquad 2s - 1 = 0$

$\boxed{s = \frac{1}{2}}, \qquad \boxed{s = \frac{1}{2}}$

Hence, 1 / 2 is a zero of the given polynomial.

Ques. 4 Find the zeroes of the quadratic polynomial $\sqrt{3}x^2 + 10x + 7\sqrt{3}$.

Sol. Let $p(x) = \sqrt{3}x^2 + 10x + 7\sqrt{3}$

For zeroes of $p(x)$, put $p(x) = 0$

$\Rightarrow \quad \sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$

$\Rightarrow \quad \sqrt{3}x^2 + 3x + 7x + 7\sqrt{3} = 0$

$\Rightarrow \quad \sqrt{3}x(x + \sqrt{3}) + 7(x + \sqrt{3}) = 0$

$\Rightarrow \quad (x + \sqrt{3})(\sqrt{3}x + 7) = 0$

$\Rightarrow \quad x + \sqrt{3} = 0, \quad \text{or} \quad \sqrt{3}x + 7 = 0$

$\boxed{x = -\sqrt{3}}, \quad \text{or} \quad \boxed{x = \frac{-7}{\sqrt{3}}}$

Hence, $\sqrt{-3}$ and $\frac{-7}{\sqrt{3}}$ are zeroes of the given quadratic polynomial.

Ques. 5 If zeroes of the polynomial $3x^2 - 2kx + 2m$ are 2 and 3, then find the value of k and m.

Sol. Given, $p(x) = 3x^2 - 2kx + 2m$ and 2 and 3 are zeroes of $p(x)$.

$\therefore \quad p(2) = 0$

$\Rightarrow \quad 3(2)^2 - 2k(2) + 2m = 0$

$\Rightarrow \quad 12 - 4k + 2m = 0$

$\Rightarrow \quad -4k + 2m = -12$

$\Rightarrow \quad 2k - m = 6 \qquad \ldots \text{(i)}$

and $\quad p(3) = 0$

$\Rightarrow \quad 3(3)^2 - 2k(3) + 2m = 0 \Rightarrow 27 - 6k + 2m = 0$

$\Rightarrow \qquad -6k + 2m = -27$

$\Rightarrow \qquad 6k - 2m = 27 \qquad \ldots$ (ii)

On multiplying by 2 in Eq (i) and subtract from Eq (ii), we get

$$\begin{array}{r} 6k - 2m = 27 \\ 4k - 2m = 12 \\ (-) \quad (+) \quad (-) \\ \hline 2k = 15 \end{array}$$

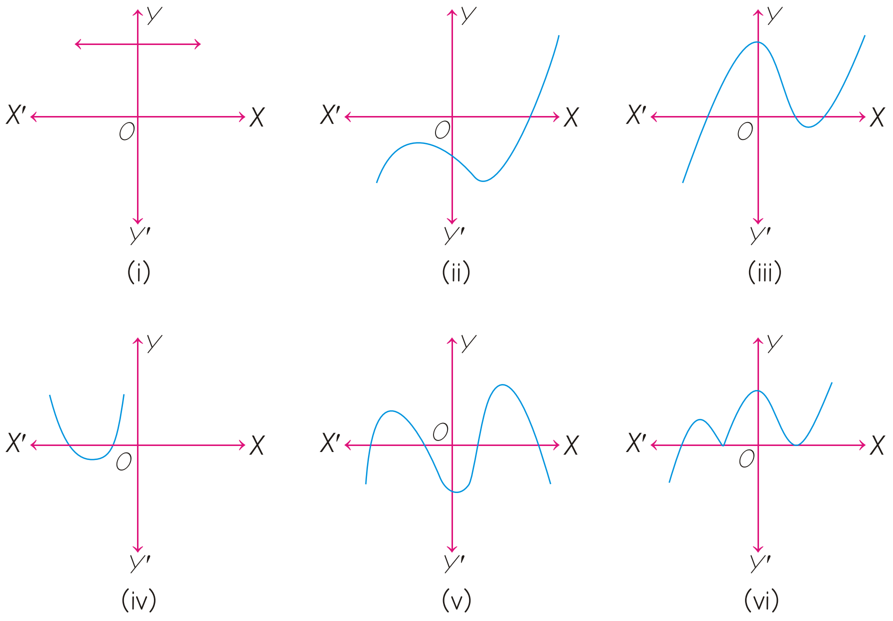
$\therefore \qquad \boxed{k = \frac{15}{2}}$

Now, on putting the value of k in Eq. (i), we get

$\Rightarrow \qquad 2\left(\frac{15}{2}\right) - m = 6 \qquad \Rightarrow \; 15 - m = 6$

$\therefore \qquad \boxed{m = 9} \qquad$ Hence, $k = \frac{15}{2}$ and $m = 9$

Ques. 6 The graphs of $y = p(x)$ are given below, for some polynomials $p(x)$. Find the number of zeroes of $p(x)$, in each case. NCERT



Sol. (i) Here, the graph of $p(x)$ does not intersect X-axis at any point.

Hence, $p(x)$ has no zero.

(ii) Here, the graph of $p(x)$ intersects the X-axis at one point.

Hence, the number of zeroes of $p(x)$ is 1.

(iii) Here, the graph of $p(x)$ intersects the X-axis at three points.

Hence, the number of zeroes of $p(x)$ is 3.

(iv) Here, the graph of $p(x)$ intersects the X-axis at two points.

Hence, the number of zeroes of $p(x)$ is 2.

(v) Here, the graph of $p(x)$ intersects the X-axis at four points.

Hence, the number of zeroes of $p(x)$ is 4.

(vi) Here, the graph of $p(x)$ intersects the X-axis at three points.

Hence, the number of zeroes of $p(x)$ is 3.

Ques. 7 Find the zeroes of the following quadratic polynomial, and verify the relationship between the zeroes and the coefficients.

(a) $4u^2 + 8u$ (b) $t^2 - 15$

Sol. (a) Let $f(u) = 4u^2 + 8u$

On putting $f(u) = 0$, we get

$$\Rightarrow \quad 4u^2 + 8u = 0$$

$$\Rightarrow \quad 4u(u + 2) = 0$$

$$4u = 0 \quad \text{or} \quad u + 2 = 0$$

$$\boxed{u = 0,} \quad \text{or} \quad \boxed{u = -2}$$

Let, $\alpha = 0, \beta = -2$

$$4u^2 + 8u + 0 \Rightarrow a = 4, b = 8, c = 0$$

$$\therefore \quad \alpha + \beta = -\frac{b}{a}$$

$$\Rightarrow \quad 0 + (-2) = -\frac{8}{4}$$

$$\Rightarrow \quad -2 = -2$$

and $$\alpha\beta = \frac{c}{a}$$

$$\Rightarrow \quad 0(-2) = \frac{0}{4}$$

$$0 = 0 \qquad \boxed{\text{verified}}$$

(b) Let $f(t) = t^2 - 15$

On putting $f(t) = 0$, we get

$$t^2 - 15 = 0$$

$$\Rightarrow \quad t^2 = 15$$

$$\Rightarrow \quad t = \pm\sqrt{15}$$

$$\Rightarrow \quad t = \sqrt{15}, -\sqrt{15}$$

Now, $\alpha = \sqrt{15}, \beta = -\sqrt{15}$

$t^2 + 0b - 15 = 0 \Rightarrow a = 1, b = 0, c = -15$ $\therefore \alpha + \beta = -\frac{b}{a}$

$\Rightarrow \sqrt{15} + (-\sqrt{15}) = -\frac{0}{1}$

$\Rightarrow 0 = 0$ and $\alpha\beta = \frac{c}{a}$

$\Rightarrow \sqrt{15} \times -\sqrt{15} = -\frac{15}{1}$

$\Rightarrow -15 = -15$ $\boxed{\text{Verified}}$

Ques. 8 If α and β are the zeroes of the polynomial $2y^2 + 7y + 5$, then find the value of $\alpha + \beta + \alpha\beta$.

Sol. We have, $2y^2 + 7y + 5$

On comparing with $ay^2 + by + c$, we get

$$a = 2, b = 7, c = 5$$

$\therefore \alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$

$\Rightarrow \alpha + \beta = -\frac{7}{2}$ and $\alpha\beta = \frac{5}{2}$

$\therefore \alpha + \beta + \alpha\beta = -\frac{7}{2} + \frac{5}{2} = -\frac{2}{2} = -1$

Ques. 9 If zeroes α and β of a polynomial $x^2 - 7x + k$ such that $\alpha - \beta = 1$, then find the value of k.

Sol. We have, $x^2 - 7x + k$

On comparing with $ax^2 + bx + c$, we get

$$a = 1, b = -7, c = k$$

$\therefore \alpha + \beta = \frac{-b}{a} = \frac{7}{1} = 7$ and $\alpha\beta = \frac{c}{a} = \frac{k}{1} = k$

Also, $\alpha - \beta = 1$ (given)

We know that $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$\Rightarrow (1)^2 = (7)^2 - 4k$

$\Rightarrow 4k = 49 - 1 \Rightarrow 4K = 48$

$\therefore \boxed{k = 12}$

Ques. 10 **If the product of the zero of the polynomial $px^2 - 6x - 6$ is 4. Find the value of p.**

Sol. Let $p(x) = px^2 - 6x - 6$

On comparing with $ax^2 + bx + c$, we get

$$a = p,\ b = -6,\ c = -6$$

$\therefore$ $$\alpha\beta = \frac{c}{a}$$

$\Rightarrow$ $$4 = -\frac{6}{p}$$

$\Rightarrow$ $$p = -\frac{6}{4}$$

$\Rightarrow$ $$\boxed{p = -\frac{3}{2}}$$

Ques. 11 **If one zero of the polynomial $3x^2 - 8x + 2k + 1$ is seven times the other, then find the value of k.**

Sol. Let $p(x) = 3x^2 - 8x + 2k + 1$

On comparing with $ax^2 + bx + c$, we get

$a = 3,\ b = -8,\ c = 2k + 1$

Let α be the one zero of the given polynomial.

$\therefore$ 7α is the other zero.

Now, $\alpha + 7\alpha = \frac{-b}{a}$ and $\alpha\beta = \frac{c}{a}$

$\Rightarrow$ $8\alpha = \frac{8}{3}$, and $\alpha \times 7\alpha = \frac{2k+1}{3}$

$\Rightarrow$ $\alpha = \frac{1}{3}$, and $7\alpha^2 = \frac{2k+1}{3}$

Now, $$7\alpha^2 = \frac{2k+1}{3}$$

$\Rightarrow$ $$7 \times \left(\frac{1}{3}\right)^2 = \frac{2k+1}{3} \quad \Rightarrow \quad 7 \times \frac{1}{9} = \frac{2k+1}{3}$$

$\Rightarrow$ $$\frac{7}{3} = 2k + 1 \quad \Rightarrow \quad 7 = 6k + 3$$

$\Rightarrow$ $$6k = 4$$

$\Rightarrow$ $$k = \frac{4}{6} = \frac{2}{3}$$

$\therefore$ $$\boxed{k = \frac{2}{3}}$$

Ques. 12 Find a quadratic polynomial whose sum and product of zeroes are $\sqrt{2}$ and $\frac{1}{3}$, respectively.

Sol. Given, sum of zeroes $= \sqrt{2}$ and product of zeroes $= \frac{1}{3}$

Then, the quadratic polynomial

$$= x^2 - (\text{sum of zeroes})x + \text{Product of zeroes}$$

$$= x^2 - \sqrt{2}x + \frac{1}{3}$$

$$= \frac{3x^2 - 3\sqrt{2}x + 1}{3}$$

We can consider $3x^2 - 3\sqrt{2}x + 1$ as required polynomial as it will also satisfy the given conditions.

Ques. 13 Find the zeroes of the polynomial $p(x) = 3x^2 - 4x - 4$. Hence, write a polynomial whose each of the zeroes is 2 more than zeroes of $p(x)$. CBSE 2025 Standard

Sol. Let $p(x) = 3x^2 - 4x - 4$

For zero of $p(x)$, put $p(x) = 0$

$\Rightarrow \quad 3x^2 - 4x - 4 = 0$

$\Rightarrow \quad 3x^2 - (6 - 2)x - 4 = 0$

$\Rightarrow \quad 3x^2 - 6x + 2x - 4 = 0$

$\Rightarrow \quad 3x(x - 2) + 2(x - 2) = 0$

$\Rightarrow \quad (x - 2)(3x + 2) = 0$

$\Rightarrow \quad x - 2 = 0$ or $3x + 2 = 0$

$\Rightarrow \quad x = 2$ or $3x = -2$

$\Rightarrow \quad \boxed{x = 2}$ or $\boxed{x = -\frac{2}{3}}$

Hence, zeroes of the given polynomial are 2 and $-\frac{2}{3}$.

Now, zeroes of the required polynomial are $\left(\frac{-2}{3} + 2\right)$ and $(2 + 2)$ i.e., $\frac{4}{3}$ and 4.

$\therefore$ Required polynomial $= x^2 - (\text{Sum of zeroes})\, x + \text{Product of zeroes}$

$$= x^2 - \left(\frac{4}{3} + 4\right)x + \left(\frac{4}{3}\right)(4) = x^2 - \frac{16}{3}x + \frac{16}{3}$$

$$= \frac{3x^2 - 16x + 16}{3}$$

We can consider $3x^2 - 16x + 16$ as required polynomial because it will satisfy the given condition.